

Eigenvalues and eigenvectors

Key Points

The characteristic equation for a matrix is

$$|M - \lambda I|$$

To find the eigenvalues for a matrix solve the characteristic equation equals zero

$$|M - \lambda I| = 0$$

To find the eigenvectors for the found eigenvalues solve

$$Mv = \lambda v$$

Basic Skills

Q1. Find x such that the following matrices are singular

a) $\begin{pmatrix} x & 5 \\ 2 & x \end{pmatrix}$ b) $\begin{pmatrix} -1+x & -3 \\ 2-x & 5 \end{pmatrix}$

Q2. Find the characteristic equation of $\begin{pmatrix} 4 & 2 & -3 \\ 5 & 9 & 2 \\ 0 & -1 & 5 \end{pmatrix}$

Q3. For matrix $M = \begin{pmatrix} 3 & 5 \\ 0 & -4 \end{pmatrix}$

a) Write an matrix for $M - \lambda I$ and find the characteristic equation

b) Show that eigenvalues are $\lambda = 3, -4$

c) Find a eigenvector v such that $Mv = 3v$

Competency Skills

Q4. Given that a matrix $M = \begin{pmatrix} -8 & -4 \\ 25 & 12 \end{pmatrix}$ has an eigenvalue of $\lambda = 2$ find the corresponding eigenvector

Q5. For the following matrices work out the eigenvalues and corresponding eigenvectors

a) $M = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$ b) $M = \begin{pmatrix} 3 & 1 \\ 5 & -1 \end{pmatrix}$ c) $M = \begin{pmatrix} 5 & 1 \\ 4 & 5 \end{pmatrix}$

Q6. For $M = \begin{pmatrix} 2 & 5 & 3 \\ 0 & 9 & -1 \\ 0 & 0 & -5 \end{pmatrix}$ find the eigenvalues and eigenvectors

Q7. Explain what eigenvectors and the corresponding eigenvalues represent for a matrix M which represents some transform

Advanced Skills

Q8. For $M = \begin{pmatrix} 2 & 3 & 0 \\ 2 & 4 & 1 \\ 0 & -3 & 2 \end{pmatrix}$ find an eigenvalue corresponding to the eigenvector $v = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$ and hence find all eigenvalues and eigenvectors

Q9. A population of chickens and foxes are modelled by coupled ODE's $\frac{dc}{dt} = 4c - 2f$, $\frac{df}{dt} = c + f$

a) Using $x = \begin{pmatrix} c \\ f \end{pmatrix}$ and the ODE matrix equation $\frac{dx}{dt} = Mx$ state the matrix M

b) Show that if $x = e^{\lambda t}v$ then $Mv = \lambda v$

c) Hence find a general solution to the coupled ODE's by finding eigenvalues and eigenvectors of the matrix M

Extension

Q10. For $M = \begin{pmatrix} 3 & 1 \\ 5 & -1 \end{pmatrix}$ show that $M^2 - 2M - 8I = 0$, relate this to the characteristic equation and explain how this equation makes it easier to calculate the inverse